



PROJECT INFORMATION

Project and customer record

Project

PROJECT NAME NH Tele Neck 007	CREATED 19.07.2026
UPDATED 19.07.2026	PROJECT NOTES -

Customer

NAME -	ADDRESS -
PHONE -	EMAIL -
NOTES -	WIND DATE -



PROJECT INFORMATION

Pickup properties and test settings

Pickup

PICKUP ID NH 9.5k43 #007	WIRE TYPE Plain Enamel
GAUGE AWG42	# WINDS 9500
POLARITY South	PHASE Negative
POLE INSULATOR Kapton Tape	PROTECTION Tissue
LEADS Waxed Pushback	START WIRE (HOT) Yellow
END WIRE (GROUND) Black	

Pickup Selections

☒ GUITAR	☒ SINGLE COIL	☒ CLOCKWISE
☐ BASS	☐ HUMBUCKER	☐ COUNTERCLOCKWISE

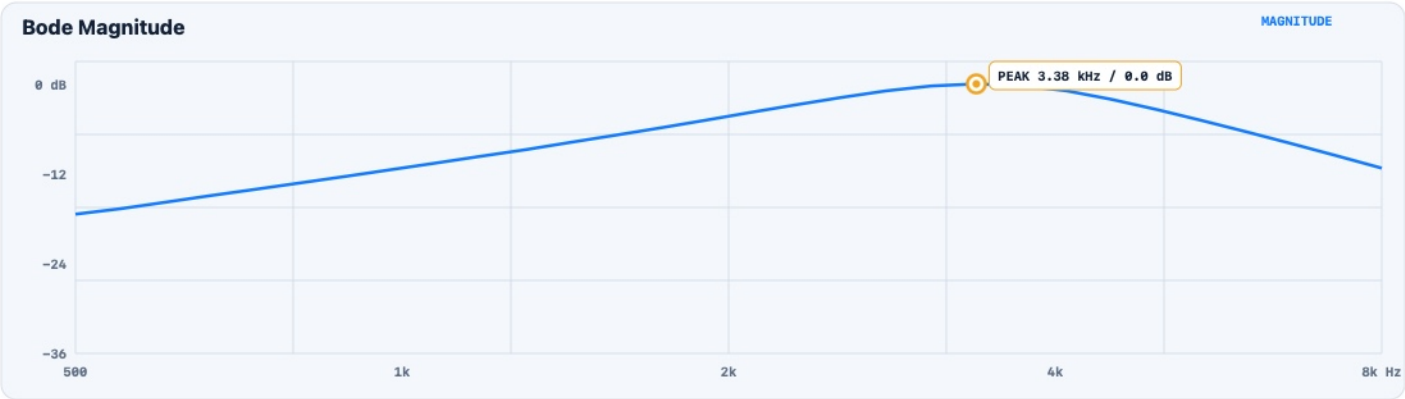
Measurement Settings

INPUT TRIM 0 dB	TEST OUTPUT LEVEL -30 dBFS
INPUT CHANNEL 1	NOISE LOW 500 Hz
NOISE HIGH 8000 Hz	NOISE DURATION 5 seconds
MANUAL TAP COUNT 5	AUTO IMPULSE COUNT 3



MEASUREMENTS

Bode response / Relative phase



Bode Response Data

PEAK RESONANCE 3.38 kHz	PEAK LEVEL -40.1 dB
500 HZ TO 8 KHZ +6.1 dB	VALID POINTS 30

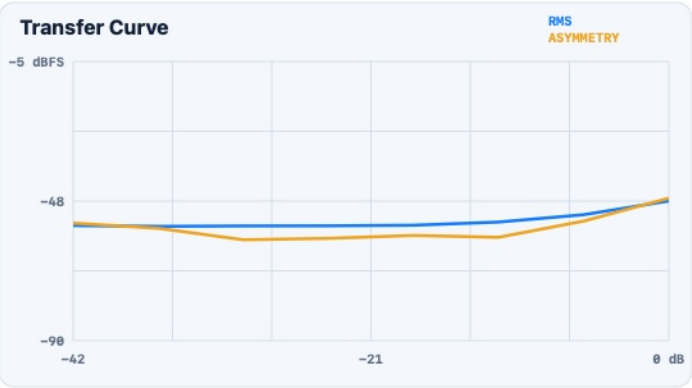
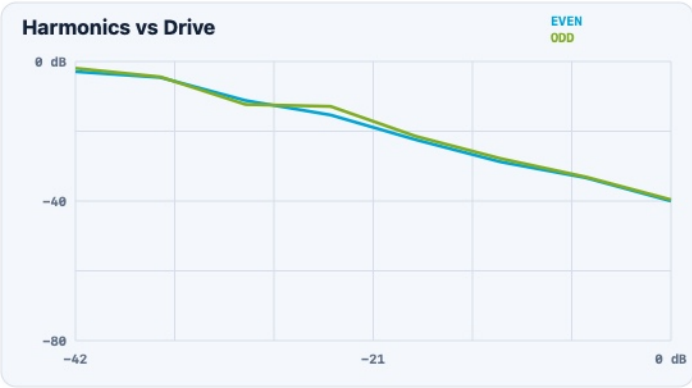
Relative Phase Data

RELATIVE PHASE NEGATIVE	CONFIDENCE 100%
POSITIVE VOTES 0 / 5	MEDIAN RESPONSE 0.0066



MEASUREMENTS

Saturation / Noise spectrum



Saturation Data

EVEN THD AT MAXIMUM
-40.0 dB
COMPRESSION
34.5 dB

ODD THD AT MAXIMUM
-39.6 dB
ASYMMETRY
0.9 dB

Noise Spectrum Data

PEAK FREQUENCY
3.16 kHz
RMS LEVEL
-46.2 dBFS
SPECTRAL CENTROID
3.94 kHz
SPECTRAL BANDWIDTH
1.45 kHz
SPECTRAL SLOPE
+0.82 dB/oct
EVEN / ODD
+1.0 dB
TRANSIENT INDEX
1.16x
MID BAND
-53.7 dB

FUNDAMENTAL
1.45 kHz
SAMPLE PEAK
-33.4 dBFS
85% ROLL-OFF
5.50 kHz
SPECTRAL FLATNESS
67.9%
THD ESTIMATE
420.3%
PEAK ASYMMETRY
+0.36 dB
LOW BAND
-140.0 dB
HIGH BAND
-42.8 dB



MEASUREMENTS

Impulse / tap



Impulse / Tap Data

CAPTURE MODE Manual	RING FREQUENCY 64 Hz
DECAY TIME / T60 ESTIMATE 0.694 s	DAMPING RATIO ZETA 0.02458
Q FROM RING 20.3	FIRST-SWING PHASE NEGATIVE
RISE TIME 5.15 ms	RESPONSES STACKED 5
NOISE REDUCTION 11.8 dB	HUM FREQUENCIES REMOVED 50 Hz, 100 Hz, 150 Hz, 250 Hz, 350 Hz



Measurement glossary

Peak resonance

Frequency of the strongest measured point in the normalized 500 Hz-8 kHz Bode sweep. It includes the driver, fixture, pickup loading, cable, and interface.

Peak level

Absolute input level at the strongest Bode point in dBFS. Use it comparatively only when every analog gain and geometric condition is unchanged.

500 Hz → 8 kHz

Level difference between the 8 kHz and 500 Hz endpoints. Positive is brighter across the measured span; negative indicates overall roll-off.

Valid points

Number of valid logarithmic sweep frequencies used in the curve. The current complete sweep contains 30 points.

Peak frequency

The strongest averaged spectral component inside the selected span. It often marks the pickup/fixture resonance, but can also expose hum or a driver peak.

Fundamental

A harmonic-score estimate of the lowest periodic component supporting upper harmonics. Broadband noise may produce low confidence, so treat this as diagnostic rather than a guaranteed musical fundamental.

RMS level

Average signal energy in dBFS across the capture. It is the best single number for repeatable overall pickup output when gain and geometry stay fixed.

Sample peak

Largest absolute digital sample in dBFS. Values near 0 dBFS warn that the interface may be clipping even when RMS looks safe.

Spectral centroid

Power-weighted center of the spectrum. Higher values generally correspond to a brighter response; compare only with the same selected span.

85% roll-off

Frequency below which 85% of selected-band spectral power lies. It summarizes how far useful energy extends toward the high end.

Spectral bandwidth

Power-weighted standard deviation around the spectral centroid. Wider bandwidth means energy is distributed across a broader frequency range.

Spectral flatness

Geometric-to-arithmetic mean power ratio. Near 100% is noise-like and flat; low values indicate concentrated peaks or resonances.

Spectral slope

Linear trend of spectral level against log2 frequency, expressed in dB per octave. Negative values indicate high-frequency roll-off.

THD estimate

Energy at harmonics 2-8 relative to the estimated fundamental. Because white noise excites every bin, treat this as a comparative screening estimate rather than calibrated distortion.

Even / odd

Ratio of even-harmonic to odd-harmonic energy around the estimated fundamental. Positive dB means even harmonics dominate; use it only for like-for-like comparisons.

Peak asymmetry

Positive-to-negative waveform peak ratio in dB. A large magnitude may reflect magnetic/electronic asymmetry, DC bias, or clipping.

Transient index

Ratio between the strongest short-time RMS block and the median block RMS. It reveals unexpected taps, handling noise, dropouts, or bursts during the noise run.

Low band

Integrated spectral level below 250 Hz, limited by the selected span. Useful for comparing hum, body, and low-frequency coupling.

Mid band

Integrated spectral level from 250 Hz to 2 kHz. This region contains much of the pickup's core musical output.



Measurement glossary

High band

Integrated spectral level above 2 kHz within the selected span. It tracks presence, resonance, and high-frequency loading.

Even THD at max

Combined 2nd, 4th, 6th, and 8th harmonics at maximum drive relative to the fundamental. Rapid even growth can indicate asymmetry, but the driver and interface can also create it.

Odd THD at max

Combined 3rd, 5th, and 7th harmonics at maximum drive relative to the fundamental. Odd growth often accompanies symmetric limiting or saturation.

Compression

Lost output growth between the lowest and highest drive steps compared with the commanded drive change. A positive value shows transfer-curve bending somewhere in the complete signal chain.

Asymmetry

Positive/negative half-cycle peak imbalance at maximum drive. Large values can reflect magnetic asymmetry, electronics, DC bias, or clipping.

Relative phase

Majority sign from five shaped-pulse responses. Positive or negative is relative to the unchanged driver face, winding, leads, and interface phase.

Confidence

Percentage of pulses agreeing with the majority sign. Below 80% is reported as indeterminate and should be repeated with better signal-to-noise.

Positive votes

How many of the five pulse responses produced a positive strongest excursion. This exposes disagreement that a single headline sign would hide.

Median response

Median absolute response amplitude across the five pulses. Very weak responses make phase votes noise-sensitive and may produce an indeterminate result.

Ring frequency

The strongest spectral component in the cleanest tap's early ring. It identifies the dominant resonant mode of the pickup and fixture; repeatability depends on identical impact and geometry.

Decay time / T60 est.

Estimated T60: the extrapolated time for the fitted logarithmic decay envelope to fall 60 dB. A noisy or multi-mode decay can make this unresolved or model-dependent.

Damping ratio ζ

Dimensionless damping ratio ζ derived as $1/(2Q)$. Smaller values mean a more lightly damped, longer-lived ring; it inherits uncertainty from the T60 fit.

Q from ring

Quality factor derived from ring frequency and T60. Higher Q means a narrower, more persistent resonance; compare only with the same fixture and electrical loading.

First-swing phase

Sign of the strongest first swing after tap onset. It is relative to tap direction and unchanged input wiring, not an absolute magnetic north/south reading.

Rise time

Time from 10% to 90% of the first peak. It describes transient onset speed and is sensitive to impact shape, bandwidth, and sample rate.